

Optimal Algorithms and Inapproximability Results for Every CSP?

by Prasad Raghavendra

Andreas Panayi

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Approximation Algorithms

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National Kapodistrian University of Athens - National Technical University of Athens

Main Meta-Theorem (informal)

In this presentation we will see that: assuming the **Unique Games Conjecture**, for every **Constraint Satisfaction Problem**:

- The optimal approximation ratio equals the integrality gap of a single natural SDP.
- That ratio is attained by solving the SDP and applying a generic rounding.

Preliminaries

Constraint Satisfaction Problem

A **Constraint Satisfaction Problem** (CSP) $\Lambda = ([q], \mathbb{P}, k)$ is specified by:

- a *finite domain* $[q] = \{0, 1, \dots, q - 1\}$ and
- a set of *predicates* $\mathbb{P}$ (predicate $P : [q]^r \rightarrow \{0, 1\}, r \leq k$).

The *arity* k of the CSP Λ is the maximum number of inputs to a predicate in $\mathbb{P}$.

Instance of a Constraint Satisfaction Problem

Every **instance** of the problem $\Lambda = [\mathcal{V}, \mathbb{P}_{\mathcal{V}}]$ consist of:

- a set of *variables* $\mathcal{V}$ and
- a set of *constraints* on them; each constraint consists of a predicate from $\mathbb{P}$ applied to a subset of variables.

The objective is to assign values from $[q]$ to the variables in $\mathcal{V}$ so as to *maximize* the number of satisfied constraints.

Example (CSP): Max-Cut

Problem Definition.

- Domain: $[q] = \{0, 1\}$.
- Predicate set: $\mathbb{P} = \{\text{NEQ}\}$ where $\text{NEQ}(a, b) = 1$ iff $a \neq b$.
 - Arity: $k = 2$.

Instance. A graph $G = (V, E)$. Define $\mathcal{V} = \{x_v \mid v \in V\}$ and $\forall (u, v) \in E$ include the constraint $(\text{NEQ}, (x_u, x_v)) \in \mathbb{P}_{\mathcal{V}}$.

Objective. Find an assignment $x : \mathcal{V} \rightarrow \{0, 1\}$ that maximizes the number of edges (u, v) with $x_u \neq x_v$.

Generalized Constraint Satisfaction Problem

A **Generalized CSP** $\Lambda = ([q], \mathbb{P}, k)$ is allowed to have a more general predicate function:

$$P : [q]^r \rightarrow [-1, 1].$$

Also, every GCSP instance $\Phi = (\mathcal{V}, \mathbb{P}_{\mathcal{V}}, W)$ has additional positive weights $W = \{w_S\}_{S \subseteq \mathcal{V}}$ satisfying $\sum_{S \subseteq \mathcal{V}, |S| \leq k} w_S = 1$.¹

Objective. Find an assignment to the variables that maximize:

$$\mathbb{E}_{S \in W}[P|_S(y|_S)] = \sum_{S, |S| \leq k} w_S P_S(y|_S)$$

¹By $S \in W$, we denote a set S chosen from the probability distribution W

More Examples

The following problems can also be written as (G)CSP:

- Max-Cut
- Max- k -Cut
- Max Di-Cut
- Max- k -SAT
- Max- k -CSP
- Max-3LIN
- Max- k -Coloring
- Multiway Cut
- Metric Labeling
- Max Acyclic Subgraph
- Label Cover
- Unique Games

and more ...

Conjecture. For every constant $1 > \delta > 0$ there exists a large enough integer R such that the following promise problem is NP-hard.

Instance. A *bipartite* Unique-Games instance

$$\Gamma = (X \cup Y, E, \Pi = \{\pi_e : [R] \rightarrow [R]\}_{e \in E}, [R]),$$

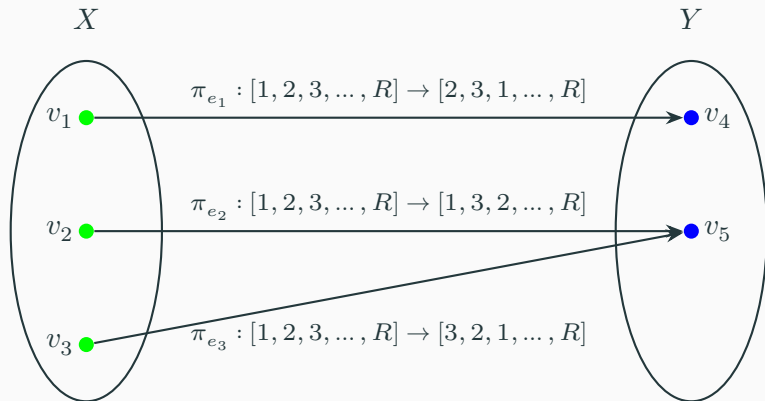
An assignment of labels $A : X \cup Y \rightarrow [R]$ is said to satisfy an edge $e = (v, w)$, if $\pi_{vw}(A(v)) = A(w)$.

Task. Distinguish between the two cases:

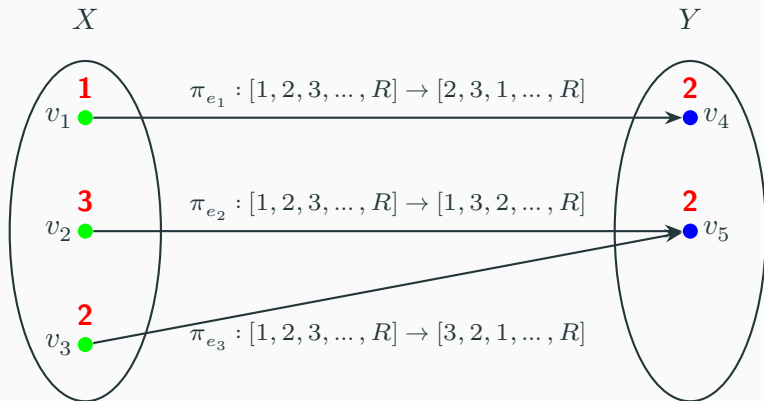
YES: Γ is $(1 - \delta)$ -satisfiable: $\exists$ assignment $A : X \cup Y \rightarrow [R]$ such that for at least $(1 - \delta)$ -fraction of vertices $v \in X$, *all* edges incident on v are satisfied.

NO: Γ is not δ -satisfiable: every labeling satisfies at most a δ -fraction of the edges of E .

Unique Games Instance: Example



Unique Games Instance: Example



Labeling $A : X \cup Y \rightarrow [1, 2, \dots, R]$ where $A(v_i)$ is equal to the red value over the vertex

A function

$$f : \{0, 1\}^R \rightarrow \{0, 1\}$$

is called a **dictator** if there exists an index $i \in [R]$ such that

$$f(x_1, \dots, x_R) = x_i.$$

Dictatorship test

A **dictatorship test** is a randomized test that, given oracle access to

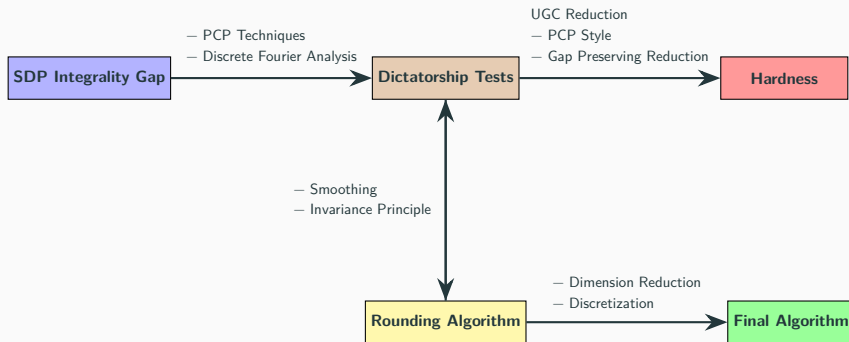
$$f : \{0, 1\}^R \rightarrow \{0, 1\},$$

makes a few queries and distinguishes between:

- **(Completeness)** f is a dictator $\Rightarrow$ the test accepts with high probability;
- **(Soundness)** f is far from every dictator $\Rightarrow$ the test accepts with low probability.

Main Results

Roadmap



Assume a GCSP $\Lambda = ([q], \mathcal{P}, k)$ and an instance $\Phi = (\mathcal{V}, \mathbb{P}_{\mathcal{V}}, W)$.

We introduce the SDP variables

$$\{ v_{(i,c)} \mid i \in \mathcal{V}, c \in [q] \}, \quad \{ X_{(S,\beta)} \mid S \subseteq \mathcal{V}, \beta \in [q]^S \}.$$

$$\text{Maximize} \quad \sum_{S \in W} w_S \left(\sum_{\beta \in [q]^S} P_S(\beta(S)) X_{(S, \beta)} \right)$$

Subject to

$$v_{(i, c)} \cdot v_{(i, c')} = 0 \quad \forall i \in [m], c \neq c' \in [q]$$

$$\sum_{c \in [q]} v_{(i, c)} = \mathbf{I} \quad \forall i \in [m]$$

$$\sum_{\substack{\beta \in [q]^S \\ \beta(s)=c, \beta(s')=c'}} X_{(S, \beta)} = v_{(s, c)} \cdot v_{(s', c')} \quad \forall S \in W, s, s' \in S, c, c' \in [q]$$

$$|\mathbf{I}|^2 = 1$$

Step 2: Dictatorship test ($DICT_{\Phi}(F)$)

Input: SDP solution for Φ , small $\varepsilon > 0$, (depends of oracle F):

1. Sample a constraint $S \in W$.
2. Draw a batch of query points from the SDP-local distribution for S .
3. Apply small independent noise to each query point to obtain $\tilde{Z}_s$.
4. Query the oracle $F : \Omega^R \rightarrow \text{Conv}(\Delta_q)$ on each $\tilde{Z}_s$.
5. Return the predicate payoff $P_S(F(\tilde{Z}_{s_1}), \dots, F(\tilde{Z}_{s_{|S|}}))$.

Step 2: Dictatorship test — completeness

Let $\text{DICT}_\Phi(F)$ be the expected payoff of the test on oracle $F : \Omega^R \rightarrow \text{Conv}(\Delta_q)$.

Completeness. If F is a dictator then

$$\text{DICT}_\Phi(F) = \text{FRAC}(\Phi) \pm o(1),$$

where the $o(1)$ -term vanishes as $\varepsilon \rightarrow 0$.

Step 3: Rounding Procedure ($Round_{\mathcal{F}}$)

Input: GCSP instance $\Phi = (\mathcal{V}, \mathbb{P}_{\mathcal{V}}, W)$, SDP solution

$\{v_{(i,c)}, X_{(S,\beta)}\}, (F : [q]^R \rightarrow \text{Conv}(\Delta_q))$

1. Sample R independent Gaussian vectors $\mathbf{g}_1, \dots, \mathbf{g}_R$.
2. For every variable $i \in [m]$:
 - (i) Compute the projections $h_{i,c}^{(j)}$ and add noise $n(\varepsilon, \mathbf{g}_j, v_{(i,c)})$:

$$h_{i,c}^{(j)} = \langle \mathbf{v}_{i,c}, \mathbf{g}_j \rangle + n$$

- (ii) Evaluate the function $\mathcal{F}$ with $h_{i,c}^{(j)}$ as inputs:

$$\mathbf{p}_i = (p_{(i,0)}, \dots, p_{(i,q-1)}) = \mathcal{F}(\{h_{i,c}^{(j)}\}_{c,j})$$

- (iii) Truncate & Normalize to ensure a valid probability distribution:

$$\tilde{\mathbf{p}}_i \leftarrow \frac{\mathbf{p}_i}{\|\mathbf{p}_i\|_1}$$

- (iv) Set label $x_i = c$ with probability $(\tilde{\mathbf{p}}_i)_c$.

Step 3: Dictatorship test — soundness

Let $F : \Omega^R \rightarrow \text{Conv}(\Delta_q)$ and $\text{Round}_F(\Phi)$ be as above.

Soundness (statement). If F is pseudorandom, then

$$\text{DICT}_\Phi(F) = \text{Round}_F(\Phi) \pm o(1),$$

and consequently

$$\text{DICT}_\Phi(F) \leq \text{INT}(\Phi) + o(1),$$

where the $o(1)$ -errors vanish as $\varepsilon \rightarrow 0$.

Step 4: Hardness Reduction ($\text{UGC} \leq \text{GCSP}_\Lambda$)

- Fix a GCSP problem Λ and a base instance $\Phi \in \Lambda$.
- Construct the *dictatorship test* DICT_Φ from the SDP solution of Φ .
- Given a Unique Games instance $\Gamma = (X \cup Y, E, \{\pi_{vw}\})$:
 - For each $w \in Y$, introduce long-code variables $\mathcal{F}^w : [q]^R \rightarrow \Delta$.
 - For each $v \in X$, define induced oracle

$$\mathcal{F}^v(z) = \mathbb{E}_{w \in N(v)}[\mathcal{F}^w(\pi_{vw}(z))].$$

- Run DICT_Φ on $\mathcal{F}^v$ to obtain queries and payoffs.
- Simulate each oracle query via neighbor queries to $\mathcal{F}^w$.
- Translate each test execution into GCSP constraints over variables

$$Y \times [q]^R.$$

Step 4: Completeness and Soundness of the Reduction

- **Completeness:** If UG has an (almost) satisfying labeling A , set dictators $F_w(z) = z_{A(w)}$. For good v , permutations align dictators so $F_v(z) = F_w(\pi_{vw}(z)) \approx z_{A(v)}$, and $DICT_\Phi$ accepts with high probability.
- **Soundness:** If $\text{val}(\Phi(\Gamma))$ is high, many F_v beat the test's soundness, hence have an influential coordinate. Since $F_v = \mathbb{E}_{w \in N(v)}[F_w \circ \pi_{vw}]$, influence transfers to neighbors as $\pi_{vw}(j)$, giving short lists $L(\cdot)$; decoding from $L(\cdot)$ satisfies many UG edges, unless $\text{val}(\Phi(\Gamma)) \leq \text{test's soundness bound}$.

Step 5: Algorithm

- **Finite cover:** Construct a *finite* family $\mathcal{F}_\kappa = \{F_1, \dots, F_l\}$ for the space of functions $F : [q]^R \rightarrow \text{Conv}(\Delta_q)$ such that for any function F , there exists $F_i \in \mathcal{F}_\kappa$ with $\|F - F_i\|_\infty \leq \kappa$.
- **Enumerate & round:** For every function $F \in \mathcal{F}_\kappa$, run the rounding scheme $\text{Round}_F(\Phi)$ and output the best assignment found.
- **Guarantee:** The algorithm outputs a solution with optimal value assuming the UGC.

Unconditional result (2-CSPs)

For Boolean 2-CSPs, the canonical SDP together with the Round procedure achieves the integrality-gap optimal approximation ratio **unconditionally** (no UGC required).

This holds because Khot–Vishnoi give explicit integrality-gap instances for the 2-CSP SDP, and the Round analysis matches those gaps directly.

Further Work and Remarks

- Consider the *basic SDP relaxation* for a fixed CSP problem.
- For every SDP solution with value ρ , there is a *polynomial-time rounding algorithm* producing an integral solution of value at least

$$\text{Gap}_{\text{SDP}}(\rho) - \varepsilon.$$

- The rounding is *universal*: the same algorithm applies to all CSPs (dependence only on arity, alphabet size, and ε).

Consequently, the integrality gap of the basic SDP is an *algorithmically achievable approximation ratio*.

- **Meta Algorithm:** A single "master" SDP and generic rounding strategy.
- **Automated Hardness:** The SDP integrality gap directly identifies the optimal approximation lower bound.
- **Pioneer Techniques:** The proof introduced groundbreaking methods, most notably the *Invariance Principle*.

References

References

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Thank You 😊
